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**Biomass-Derived Carbon Fiber Aerogel as Binder-Free Electrode for
High-Rate Supercapacitor**

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Abstract

A flexible carbon fiber aerogel with a very high surface area for supercapacitor application is reported by carbonization and chemical activation of low-cost natural cotton with KOH. The carbon fibers in the aerogel present as twisted and tubular structure. Depending on the amount of KOH used in the activation process, the specific surface area of aerogels ranges from 1536 to 2436 m² g⁻¹, while their electrical conductivities remains ~860 S m⁻¹. In spite of pore size in the range of 1.0–4.0 nm and pore volume mainly contributed by micropores, the carbon aerogel exhibits a high specific capacitance of 283 F g⁻¹ (1 A g⁻¹) in 6 M KOH aqueous electrolyte, and retains a high capacitance retention of 224 F g⁻¹ at current density up to 100 A g⁻¹. Importantly, a symmetric capacitor built with the aerogel electrodes exhibits a rather small time constant (0.56 s). The superior capacitive performance of aCF electrode is closely related to its distinct structural advantage. The tubular carbon fibers with several millimeter in length offer ultra-long electronic and ionic pathway, while plenty of nanopores on the fiber walls created by KOH activation enables fast ion transport across the walls. Our results demonstrate that capacitive performance of the traditional microporous carbon, which is characterized by poor ion kinetics, can be significantly enhanced by properly engineering the electrode architecture.

INTRODUCTION

Supercapacitors, as one of promising energy storage devices, have attracted tremendous attentions over the past decades because they can offer ultrahigh power density, fast charge-discharge rate, exceptionally long cycling life and high energy density which is even comparable to battery.¹⁻⁴ According to the energy storage mechanism, supercapacitors can be classified into electrochemical double layer capacitor (EDLC) and pseudocapacitor. The former stores energy through fast electrostatic adsorption of electrolyte ions at the electrode surface, while the later relies on reversible Faradaic redox reactions occurring at the electrolyte/electrode interface to store energy.^{5, 6}

The rapid development of various flexible and portable electronic devices has triggered significant research efforts on lightweight, low-cost and environmental friendly electrode materials.⁷⁻⁹ As one of ultralight materials with many interesting properties, carbon aerogels have received ever-increasing attentions over the past several years because of their low mass densities, high surface areas and excellent chemical stabilities. Particularly, the 3D interconnected network with open pores in the aerogels enables easy access of guest ions/molecules into the interior, making carbon aerogels one of very attractive candidates with potential applications in environment remediation and energy storage and conversion.¹⁰⁻¹⁴ The intrinsically high surface area is one of key considerations for aerogel applications. Graphene aerogels can be easily formed by self-assembly of two-dimensional graphene oxides sheets under different hydrothermal conditions.^{13, 15, 16} The strong sheet-to-sheet

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4 interactions during self-assembly and the physical cross-linking between neighboring
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6 sheets significantly reduce their accessible surface areas, thus limiting their wide
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8 applications. Although chemical activation has been applied to improve the graphene
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10 aerogel surface area,^{17, 18} this treatment was usually accompanied by dramatic
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12 reduction of the mechanical strength and flexibility. Therefore, developing other type
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14 of carbon aerogels with large surface area and good mechanical flexibility is highly
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16 desirable for portable and lightweight energy storage devices.^{7, 10, 19, 20}
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21 Carbonization of biomass represents a facile yet cost-effective route toward
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23 scalable production of porous carbon for utilization in electrochemical energy storage
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25 and conversion.²¹⁻²⁶ Cotton is abundant in nature and has been widely used for textile
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27 and clothing. Cotton can be regarded as a cellulose aerogel as it mainly consists of
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29 millimeter-scale cellulose fibers that are inter-entangled together. Carbonization of
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31 cotton by simple thermal annealing can convert the cellulose aerogel into fibrous
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33 aerogel, which not only possess the high electronic conductivity, but also maintains
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35 the mechanical flexibility and macroscopic morphology of the starting cotton,²⁷ thus
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37 showing promising electrode materials for wearable energy storage devices.²⁸ Despite
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39 of their outstanding potentials, the present carbon aerogels derived from natural cotton
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41 suffer from either low surface area or poor mechanical strength,²⁹⁻³¹ which largely
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43 hinders their applications as flexible electrodes or as robust scaffolds for
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45 electrochemical energy storage devices.
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54 Herein, we report a carbon fiber aerogel with both high surface area and good
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56 mechanical flexibility prepared by carbonization and chemical activation of natural
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cotton using KOH. As illustrated in Scheme 1, the helical and tubular carbon fibers with several millimeter in length serves as ultra-long superhighways for rapid migration of both ions and electrons along its one-dimensional (1D) backbones, while plenty of nanopores on the fiber wall created by KOH activation offers large surface area for efficient charge storage. Consequently, in spite of pore size in the range of 1.0–4.0 nm and pore volume dominantly contributed by micropores, the resulting carbon fibers deliver a high specific capacitance of 283 F g^{-1} at 1 A g^{-1} , and exhibit a remarkable capacitance retention of 79% (224 F g^{-1}) at very large current density of 100 A g^{-1} . Moreover, a capacitor built by the aerogel also exhibit a rapid frequency response with a rather small relaxation time constant of 0.56 s, which is an order of magnitude lower than those of conventional activated carbon-based capacitor. The excellent capacitive behaviors displayed by the micropore-based carbon fiber aerogel clearly demonstrate that by properly engineering the electrode architecture, both high-capacitance and high-rate can be simultaneously achieved for microporous carbon-based electrode materials.

EXPERIMENTAL SECTION

SAMPLES PREPARATION

The commercially available absorbent cotton was applied as starting materials. Heat treatment of 2.0 g cotton at 800°C in flowing N_2 for 90 min converted them into black product which was denoted as CF. The CF was subsequently subjected to an activation process using KOH as the activation agent. In a typical activation process,

CF of 0.2 g was soaked into 2.0 M KOH aqueous solution of different volumes to get a mass ratio of KOH/CF varying from 4 to 7. After evaporating the excessive water at 120 °C, the KOH/CF mixture was activated at 800 °C for 90 min with a ramp rate of 5 °C min⁻¹. After cooled down to room temperature, the sample was collected and washed with 2.0 M HCl and copious water until the pH of the filtrate reached 7. The collected sample was denoted as aCF-*x*, with *x* representing mass ratio of KOH/CF in the activation step.

CHARACTERIZATION METHODS

The morphologies of the samples were examined by field-emission scanning electron microscopy (FESEM) on SU8020. The microstructures of the samples were observed on Tecnai G2 F20 S-Twin Field-emission transmission electron microscopy (FETEM) operated at an acceleration voltage of 200 kV. Nitrogen adsorption measurement was performed at 77 K on a micromeritics ASAP 2420 analyzer. Samples were degassed at 180 °C for 6 h prior to the measurement. The specific surface area (SSA) of the samples was determined according to the Brunauer-Emmett-Teller (BET) method using the adsorption data in the relative pressure (P/P_o) range of 0.05–0.2. Total pore volumes were estimated at $P/P_o = 0.99$. The pore size distribution (PSD) was analyzed using a nonlocal density functional theory (NLDFIT) model assuming the cylinder pore geometry from the adsorption data. X-ray photoelectron spectroscopy (XPS) spectra were collected on an AXIS ULTRA spectrometer (Kratos Analytical) using a monochromatized Al Ka X-ray source (1486.71 eV). Raman spectra were measured on a Renishaw inVia Raman microscope with an excitation wavelength of

532 nm. The electrical conductivity of samples was tested by using a standard four-probe method. Samples were pressured into a pellet under a pressure of 2.0 Mpa.

ELECTROCHEMICAL MEASUREMENTS

The electrochemical performances of the electrode materials were characterized by cyclic voltammetry (CV), galvanostatic charge-discharge and electrochemical impedance spectroscopy (EIS) on a Gamary Reference 3000 electrochemical workstation with 6.0 M KOH as aqueous electrolyte. The working electrodes were prepared by fixing ~1.0 mg of aCF between two pieces of nickel foam (1.0 cm²) under a pressure of 100 kPa. Both three-electrode cell and two-electrode system were used to characterize electrode electrocapacitive performance. In a three-electrode cell, a Pt foil and a Ag/AgCl electrode were applied as the counter and reference electrodes, respectively. The specific capacitance, C (F g⁻¹) of the electrode material was calculated from the galvanostatic discharge curves according to the following equation: $C = I \times \Delta t / (\Delta V \times m)$, where I is the discharge current (A), Δt is the discharge time (s), ΔV is the voltage change (V) excluding voltage drop (IR drop) in the discharge process, and m is the mass of the active material (g).^{32, 33} In a two-electrode cell, a Swagelock-type capacitor was configured with Celgard®3501 as the separator. The specific capacitance was calculated according to the equation: $C_s = 4 \times I \times \Delta t / (\Delta V \times m)$, where I , Δt and ΔV are discharge current (A), discharge time (s) and the voltage change (V) excluding the IR drop, respectively. Whereas, m is the total mass of the active material (g) in two electrodes.^{34, 35} The ion kinetics within an electrode material were investigated by electrochemical impedance spectroscopy (EIS) on a capacitor

with an amplitude of 10 mV at the frequency range of 0.01 Hz to 100 kHz. The real part $C'(\omega)$ and the imaginary part $C''(\omega)$ of the capacitance were extracted from the impedance data according to the equation:

$$C'(\omega) = \frac{-Z''(\omega)}{2\omega|Z(\omega)|^2} \quad C''(\omega) = \frac{-Z'(\omega)}{2\omega|Z(\omega)|^2}$$

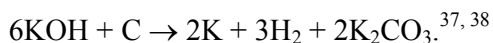
Where $Z(\omega)$ is the complex impedance, $Z'(\omega)$ and $Z''(\omega)$ are the real and imaginary part of the complex impedance, respectively.³⁶

RESULTS AND DISCUSSION

The natural cotton is composed of curled cellulose fibers with several millimeter in length and 10–20 μm in diameter (Figure S1). The thermogravimetric analysis of natural cotton in flowing N_2 (not shown) reveals that ~8 wt% residua was obtained at 800 °C. Consequently, we carbonized the starting cotton at this temperature to convert the cellulose fibers into carbon fibers. As shown in Figure 1a, the carbon fibers resemble the pristine cellulose fibers but with distinct reduction in both length and diameter due to the substantial volume contraction during carbonization process. The obtained CF was subsequently subjected to chemical activation to improve its surface area. Figure 1b shows the digital photos of aCF-6 which was prepared with a mass ratio of $\text{KOH}/\text{CF} = 6$. Like the loosely cotton, the aCF-6 aerogel can sustain large deformation under manual compression and maintain a good structure integration without evident damage (Figure 1b). Figure S2 shows the stress-strain curves of aCF-6 sample tested by three successive strain experiments under stress. As aCF-6 is highly loose, it can sustain large deformation of 65% under a stress of 35 kPa. After stress release, the aerogel can recover to its initial state, as verified by the nearly

identical stress-strain curves in the second and third stress-strain experiment (Figure S2). The highly compression-recovery property of the aCF-6 sample is vital for its potential use in flexible energy storage devices. SEM image shows that the aCF-6 is composed of numerous carbon fibers analogous to the original cellulose fibers (Figure 1c and Figure S3). A close inspection reveals that most of the fibers presents as helical and tubular structure with wall thickness ranging from 1.5 to 2.0 μm and tubular diameter varying from 5.0 to 7.0 μm (Figure 1d, e). A four-probe conductivity measurement on the pressed pellet shows that aCF-6 has an electrical conductivity of $\sim 860 \text{ S m}^{-1}$ (Figure 1f), indicative of an intrinsically high electron transfer capability.

Chemical activation of carbon materials with KOH is suggested to proceed as



Reaction of carbon with KOH at high temperature eliminates carbon atoms and leaves behind numerous vacancies on the product. Generation of porosity on the fiber wall was verified by TEM. As shown in Figure 2, highly porous network can be clearly seen in the low-magnification images (Figure 2a, 2b). Like other biomass-derived carbon products by KOH activation,^{39, 40} the aCF-6 is mainly made up of disordered and wormlike micropores with pore size around 2.0 nm (Figure 2c, 2d). This observation suggests the effectiveness of chemical activation in creating porosity on the fibers walls. The phase structures of aCF products were characterized by XRD and Raman spectra. As shown in Figure 3a, two broad and low-intensity diffraction peaks at 23.7° and 41.1° are observed for CF sample, which can be indexed to (002) and (101) spacing of graphitic carbon (JCPDS No.75-1621). The weak peaks suggest a

low graphitization of CF sample. After chemical activation, these two peaks become weaker and slightly shifts toward lower angles with a higher KOH/CF ratio, indicating a less ordered graphitic structure of aCF possibly due to harsh activation by KOH treatment. Figure 3b presents the Raman spectra of CF and aCF. The D bands at 1340 cm^{-1} are related to the disordered sp^3 carbon atoms, whereas G bands at 1593 cm^{-1} correspond to sp^2 -hybridized carbon atoms in the graphitic layers.³⁸ The relatively higher $I_{\text{D}}/I_{\text{G}}$ of aCF (0.94–0.99) with respect to that of pristine CF (0.91) (Table 1) suggests more defect sites were introduced in aCF during the activation process. This variation matches well with XRD results and accounts for the relatively low electrical conductivity of aCF (Figure 1f). The surface chemistry of CF and aCF probed by XPS give only C 1s and O 1s peaks (Figure S4). The atomic percentages of O in the CF and aCF products vary from 6.9 to 8.1% based on the XPS quantitative analysis (Table 1).

The highly porous texture of aCF was further verified by N_2 adsorption/desorption measurement. As shown in Figure 3c, the very low N_2 uptake of CF reveals its nearly non-porous structure. In contrast, all aCF products display significantly enhanced N_2 uptake and obvious capillary condensation at relative pressure (P/P_0) below 0.1, which is ascribed to type I due to adsorption of N_2 in abundant micropores. Moreover, the continuous increase of the adsorption capacity with the KOH/CF ratio implies more micropores were created in aCF. Figure 3d shows the pore size distribution calculated from NLDFT. The pore size in aCF is mainly distributed in the range of 1.0–4.0 nm, with a slight enlargement at a higher

KOH/CF ratio. This pore size enlargement is likely caused by merging neighboring small micropores into larger ones due to the excessive KOH activation. Nevertheless, the progressive etching of carbon atoms with KOH results in a continuous increase of both the SSA and pore volume (Table 1). For instance, aCF-6 exhibits a SSA of 2307 $\text{m}^2 \text{g}^{-1}$ and a pore volume of 1.18 $\text{cm}^3 \text{g}^{-1}$, which are over twenty times higher than those of pristine CF. By analyzing the plots of cumulative pore volume vs pore size (Figure S5), the ratio of micropores volume (V_{mic}) to the total pore volume (V_{t}) for aCF-4 is 75.3%, which gradually decreases to 65.3% for aCF-6 (Table 1). The decrease of $V_{\text{mic}}/V_{\text{t}}$ at a higher KOH/CF ratio is in good consistence with the pore size enlargement, thus supporting a possible pore merging during activation process.

The aCF was directly applied as a binder-free supercapacitor electrode and their electrochemical performances in 6 M KOH were tested by a three-electrode cell with 6.0 M KOH as aqueous electrolyte. Figure 4a compares the galvanostatic charge–discharge profiles of CF and aCF electrodes. All electrodes exhibit nearly symmetric charging and discharging profiles, suggesting an ion adsorption charge storage mechanism. The specific capacitances of different electrodes at current density of 1.0 A g^{-1} were summarized in Table 1. It is seen that CF electrode shows a low capacitance of 122 F g^{-1} because of its non-porous structure. Whereas, all aCF electrodes exhibit gradually increased specific capacitances due to the increased SSA. In particular, aCF-6 electrode reaches a maximal capacitance of 283 F g^{-1} at 1 A g^{-1} . It is noted that aCF-7 electrode has a higher SSA but delivers a lower capacitance. According to pore volume vs pore size analysis (Figure S5), the micropores volume is

measured to be 0.77 and 0.80 cm³ g⁻¹ for aCF-6 and aCF-7, respectively. This analysis suggests that some of micropores in aCF-7 is actually not accessed by K⁺ and/or OH⁻ although these micropores contribute to a high SSA. In addition to a high capacitance, aCF-6 electrode also exhibits a outstanding capacitance retention of 224 F g⁻¹ upon increasing the current density to 100 A g⁻¹ (Figure 4b and Table 1). It is noted that the capacitive performance of aCF-6 reported herein is superior to those of previous fiber-based electrode, including cotton-derived carbon electrode,^{20, 31} heteroatom-doped carbon nanofibers,^{41, 42} porous carbon-graphene composite electrodes,⁴³ and other biomass-derived carbon electrodes.⁴⁴⁻⁴⁶ The high capacitance of aCF-6, along with its unusual capacitance retention (79%) over a wide current density range (1–100 A g⁻¹) clearly demonstrates that the abundant nanopores randomly distributed on the fiber walls not only offer large ion-accessible surface area for efficient charge storage, but also serves as ion pathways allowing fast diffusion across the tubular fiber walls.

In order to evaluate the capacitive performances of aCF electrode for real supercapacitor, a symmetric capacitor was built by using aCF-6 as the electrode without use of any binder and conductive additive. Figure 5a-e present the CV profiles of aCF-6-based device at different scan rates. It is interesting that the aCF-6-based capacitor remains a good rectangular CV profiles at both low and high scan rate, demonstrating aCF-6 electrode can serve as one of promising electrodes for high-rate supercapacitor. It is suggested that the charge storage mechanism can be evaluated by the following equation:

$$i = av^b$$

where i is the peak current (mA), v is the scan rate (mV s^{-1}), a and b are coefficients.⁴⁷ The value $b = 0.5$ indicates a semi-infinite diffusion-controlled process, while $b = 1$ represents a capacitive behavior.^{48, 49} In order to reveal the charge storage mechanism of aCF-6 device in 6 M KOH, we extracted the discharge current densities at 0.5 V from the CV curves and plotted the $\log(\text{discharge current density (A g}^{-1}\text{)})$ vs $\log(\text{scan rate (mV s}^{-1}\text{)})$ in Figure 5f. By fitting the plot over a wide range of scan rates from 100 to 1000 mV s^{-1} , the coefficient $b = 1.01$ was obtained. This experimental result unambiguously reveals that energy storage in aCF-6-based device is achieved by fast and reversible electrostatic ion adsorption at the electrode/electrolyte interface.

Figure 6a shows the galvanostatic charge–discharge profiles of aCF-6-based capacitor. The almost symmetric triangular-like shapes of charge-discharge curves further confirms the electronic double-layer energy storage mechanism. It is noted that even at high current density of 50 A g^{-1} , the discharge curves of aCF-6-based capacitor only exhibits a rather small voltage drop (IR drop) (0.135 V) (Figure S6), suggesting a rather low internal resistance of aCF-6-based device. As a key component of a supercapacitor, the property of the electrode materials, including the electronic conductivity and ion transport capability, predominately determines the internal resistance of a device. As discussed above, aCF-6 made up of tubular carbon fibers with a few millimeter in length can serve as ultra-long 1D pathway for both ion and electron transports (Scheme 1). Meanwhile, plenty of nanopores on the fiber walls enables easy access of ions to the electrode surface. The distinct structural advantage

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4 results in aCF-6 to exhibit high-rate capability. As shown in Figure 6b, upon
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6 increasing the current density from 0.1 to 50 A g⁻¹, 77% initial capacitance is still
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8 retained (193 F g⁻¹).
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11 The fast ion diffusion and nearly ideal capacitive behavior of the aCF-6-based
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13 capacitor are also manifested in its EIS. The equivalent circuit diagram used for fitting
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15 the EIS (inset of Figure 6c) consists of solution resistance (R_s), the charge transfer
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17 resistance (R_{ct}), a pseudocapacitive element (C_p) from surface oxygen-containing
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19 groups and the constant phase element (CPE) to interpret the double layer capacitance.
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21 The Nyquist plot of aCF-6-based capacitor consists of a vertical line at low frequency
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23 region and a semicircle at high frequency region (Figure 6c and inset). The nearly
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25 vertical line further verifies the ideal capacitive behavior of the device. By fitting the
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27 EIS using the equivalent circuit diagram, $R_s = 0.23$ ohm and $R_{ct} = 0.68$ ohm were
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29 determined. The rather low R_s and R_{ct} imply a good ion conductivity and a fast
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31 electron transfer process of the device. The knee frequency which denotes the
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33 maximum frequency below which the capacitive behavior is dominate,⁵⁰ was
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35 measured to be 16.8 Hz (inset in Figure 6c). Note this frequency is much higher than
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37 3.0 Hz of graphene aerogel-based capacitor.⁵¹ Figure 6d shows the real (C') and
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39 imaginary (C'') plots of the aCF-6-based capacitors. The imaginary part of
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41 capacitance goes through a maximum at frequency f_0 (1.78 Hz), which defines the
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43 relaxation time constant τ_0 ($= 1/f_0$) and marks the point where the resistive and
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45 capacitive impedance are equal.⁵⁰ The $\tau_0 = 0.56$ s for aCF-6-based capacitor is even
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47 comparable to that of capacitors with carbon nanomesh and holey graphene foam
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electrode (0.46 and 0.49 s),^{35, 52} but rather smaller than those with activated carbon electrode (1.2 and 2.45 s)^{38, 53} and chemically activated graphene electrode (0.73 and 1.67 s)^{17, 54} In sharp contrast to conventional activated carbon electrodes which have a high SSA but are characterized by poor rate performance due to slow ion kinetics within narrow and long channels (typical of tens of micrometers), the rapid frequency response of the aCF-6 capacitor reported herein clearly demonstrates that the tubular structure of aCF-6 electrode with plenty of nanopore on the conductive fiber wall can potentially shorten the ion diffusion lengths, promote the electron migration, thus enabling a fast ion and electron transport within the interior of electrode.

Apart from the high-rate capability, the aCF-6-based capacitor also exhibits a remarkably high cycling stability. Figure 7 shows the cycling performance of the aCF-6 capacitor tested by continuous charge–discharge at current density of 4 A g⁻¹ for 20000 cycles. The capacitance in the initial 1500 cycles is found to gradually decrease, probably due to the filling and wetting the ink-bottom pores to achieve a stable charge-discharge state.³ In the following 15000 cycles, the capacitance recovers gradually and only ~3% capacitance decay is found. The cycling stability of the electrode is also seen from the nearly identical CV and charge-discharge profiles recorded at different cycling stages (inset in Figure 7). More importantly, results from the Nyquist plots also indicates that both R_s and R_{ct} have negligible changes during the whole cycling process. This observation clearly demonstrates that aCF-6 electrode can remain nearly unchanged ionic and electronic transport properties during the long-term cycling. These intrinsic properties are particularly important for real

application of aCF-6 electrode in electrochemical energy storage.

CONCLUSIONS

Carbon fiber aerogel with a high surface area have been prepared by carbonization and chemical activation of natural cotton. The obtained aCF products remain the highly elastic properties of starting cotton while possessing a high surface area and good electrical conductivity. In sharp contrast to the traditional conception that activated carbons with majority of micropores usually show poor ion kinetic, the aCF-6 electrode reported herein has the capability to offer a high specific capacitance, high-rate performance and excellent cycling stability although it is dominated by micropores. This work clearly show that properly engineering the electrode architecture is the key to realize the full potential of microporous carbon in electrochemical energy storage. The compression-recovery property also makes the carbon fiber aerogel one of ideal scaffolds to accommodate various pseudoactive electrodes for flexible energy storage devices.

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Supporting Information Available SEM image of cotton and aCF-6 with different magnifications, stress-strain curves of aCF-6 sample, survey XPS spectra, cumulative pore volume vs pore size of CF and aCF, and IR drop of aCF-6-based capacitor at current density of 20 and 50 A g⁻¹. This information is available free of charge via the Internet at <http://pubs.acs.org>

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Captions for Figures and Table

Scheme 1. Schematic illustrating the fabrication steps and ions/electrons transport behavior in aCF electrode.

Figure 1. SEM image of CF (a) and digital photo of aCF-6 before and after manual compression (b). SEM images of aCF-6 (c-e) with different magnifications, and electrical conductivity of CF and aCF measured by a standard four-probe method (f).

Figure 2. TEM images of aCF-6 with low (a, b) and high (c, d) resolutions.

Figure 3. Powder XRD pattern (a), Raman spectra (b), N₂ adsorption isotherms (c), and corresponding NLDFT pore size distribution (d) of CF and aCF samples.

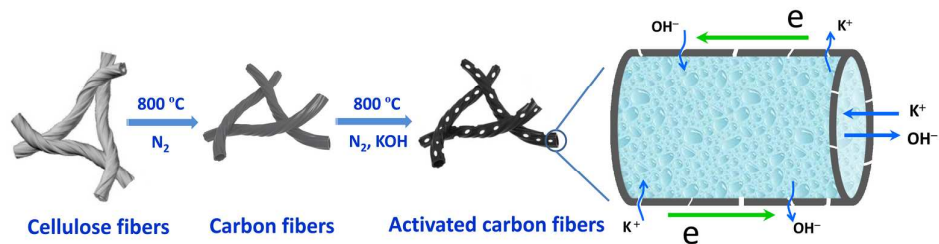
Table 1. Structure parameters and capacitive performance of CF and aCF electrodes.

Figure 4. Galvanostatic charge–discharge curves (a), and capacitance retention (b) of CF and aCF electrodes in a three-electrode cell.

Figure 5. CV profiles of aCF-6-based capacitor over a wide range of scan rates from 100 to 1000 mV s^{−1} (a-e), and the dependence of the discharge current density on the scan rates (f). *b*-value was determined by fitting a log(discharge current density)-log(scan rate) plot. The discharge current densities were extracted at 0.5 V from the CV curves of different scan rates.

Figure 6. Capacitive performances of aCF-6-based capacitor in 6.0 M KOH electrolyte. (a) galvanostatic charge–discharge curves, (b) capacitance retention at different current densities, (c) Nyquist plots of aCF-6-based capacitors with high-frequency region and equivalent circuit diagram given in inset, (d) Evolution of real (*C'*) and imaginary (*C''*) parts of specific capacitance of aCF-6-based capacitors as a function of frequency.

Figure 7. (a) Cycling performance of aCF-6 capacitor at current density of 4 A g^{−1} with inset showing CV profiles at 20 mV s^{−1}, galvanostatic charge–discharge curves, and Nyquist plots recorded at different cycling stages.



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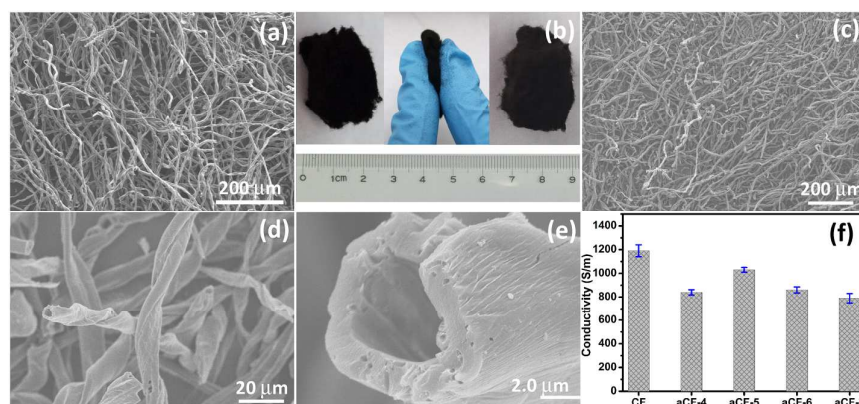


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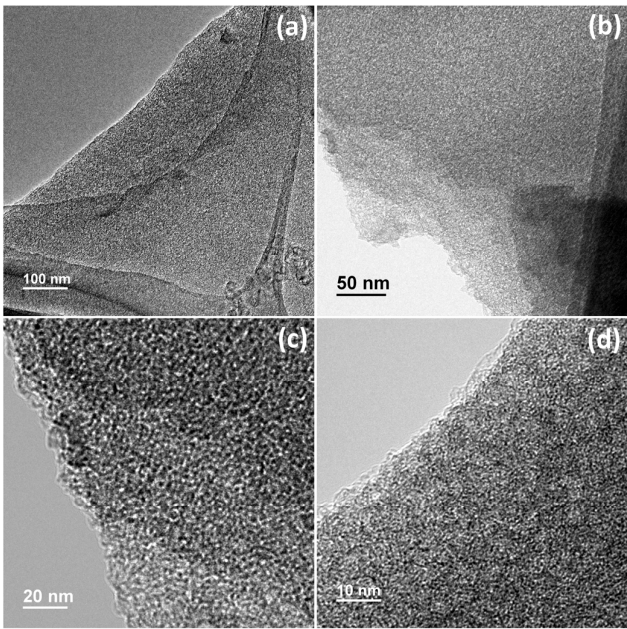


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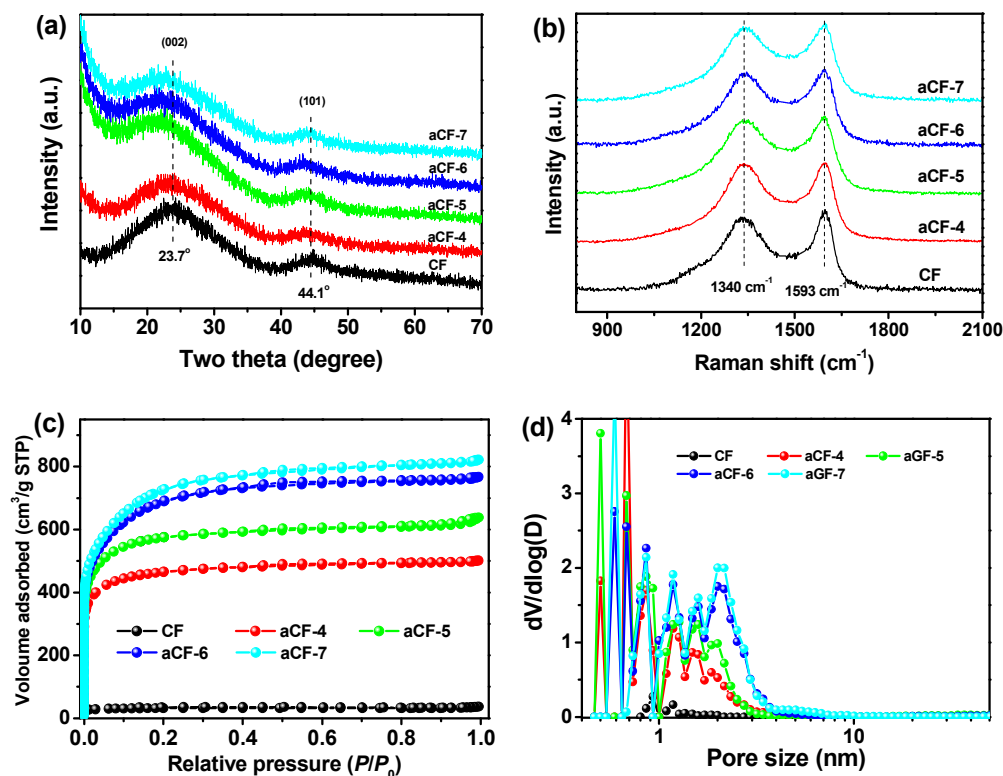


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Table 1. Structure parameters and capacitive performance of CF and aCF electrodes.

Samples	Element composition (at%)		SSA (m ² g ⁻¹)	V_t^a (cm ³ g ⁻¹)	V_{mic}/V_t^b (×100%)	I_D/I_G	Specific capacitance (F g ⁻¹) ^c	Capacitance retention (%) (100 A g ⁻¹)
	C	O						
CF	93.1	6.9	104	0.06	–	0.91	122	32
aCF-4	91.7	8.1	1536	0.77	75.3	0.99	262	71
aCF-5	92.2	7.8	1898	0.98	73.5	0.94	283	67
aCF-6	92.6	7.4	2307	1.18	65.3	0.97	283	79
aCF-7	92.1	7.9	2436	1.27	63.0	0.97	267	76

^a Total pore volume measured at relative pressure of 0.99. ^b Micropore volume (pore size < 2 nm) analyzed from NLDFT. ^c Specific capacitance measured at current density of 1.0 A g⁻¹ in three-electrode cell with 6.0 M KOH as aqueous electrolyte.

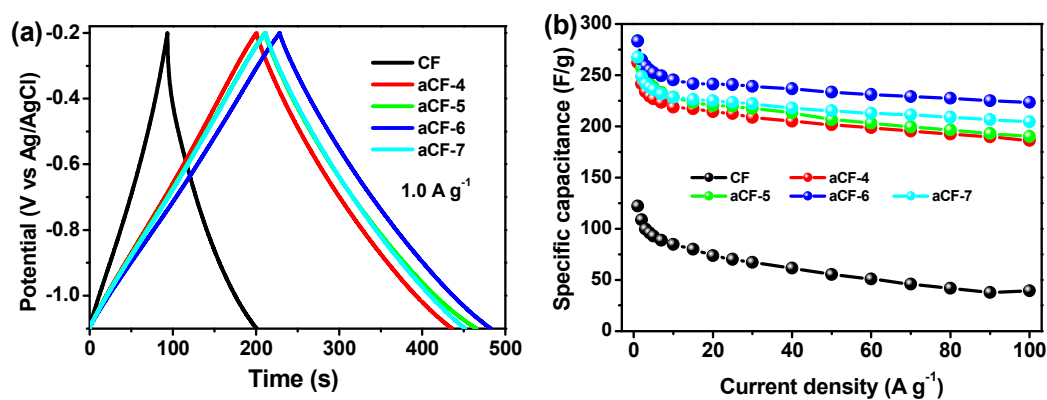


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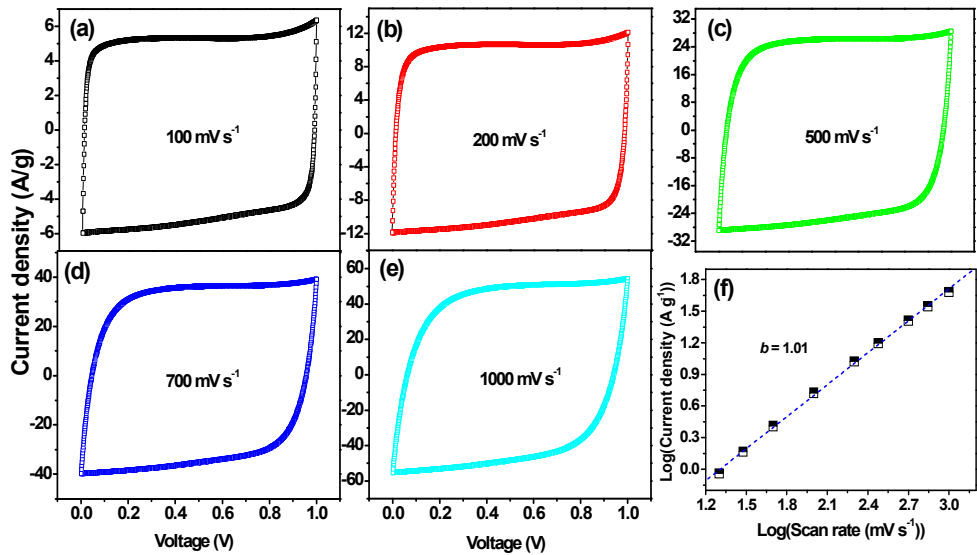


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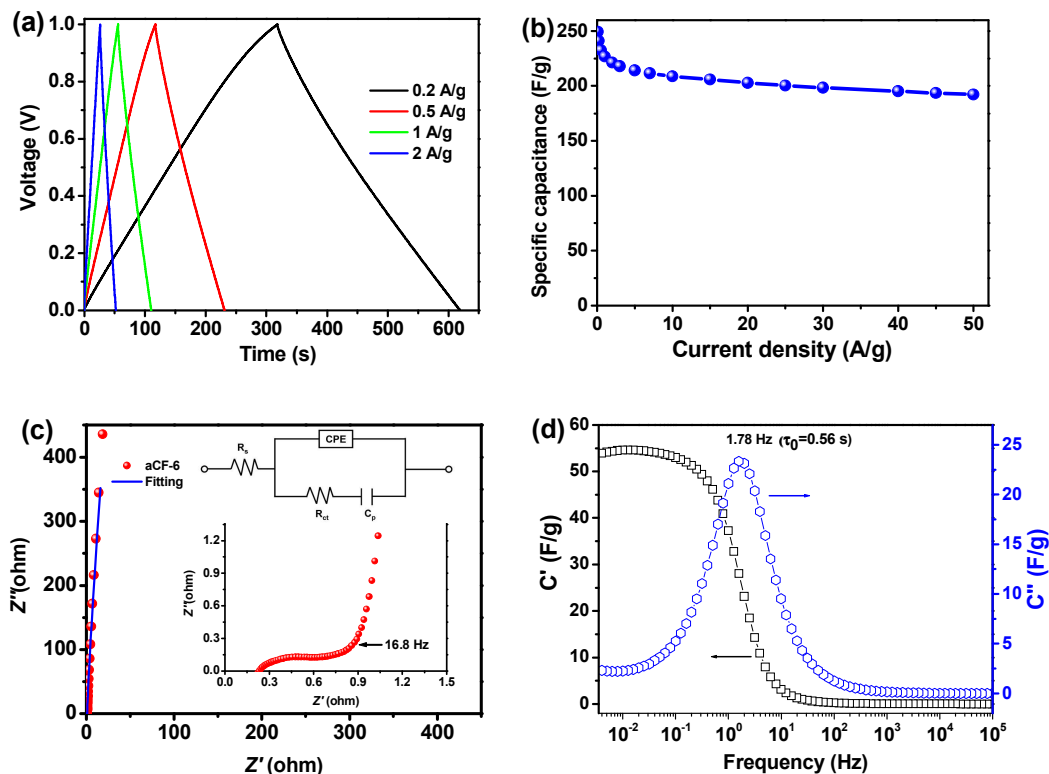


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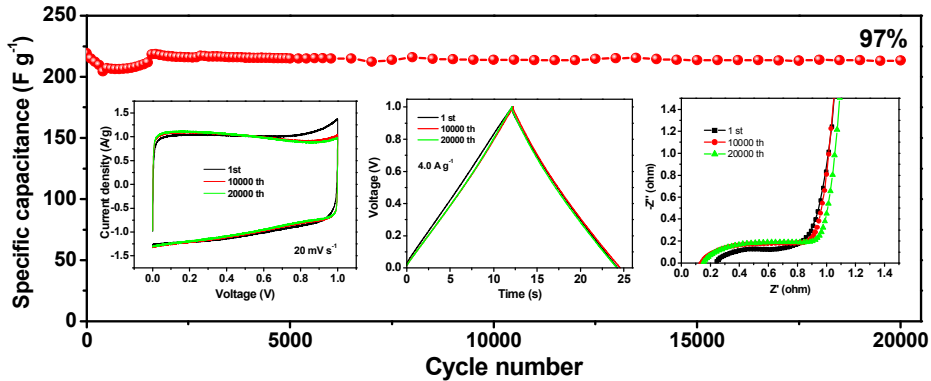


Figure 7. (a) Cycling performance of aCF-6 capacitor at current density of 4 A g⁻¹ with inset showing CV profiles at 20 mV s⁻¹, galvanostatic charge–discharge curves, and Nyquist plots recorded at different cycling stages.

Table of Content Graphic

Biomass-Derived Carbon Fiber Aerogel as Binder-Free Electrode for High-Rate Supercapacitor

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